

Aldicarb-induced Paralysis Assay to Determine Defects in Synaptic Transmission in *Caenorhabditis elegans*.

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Aldicarb treatment causes an accumulation of acetylcholine in the synaptic cleft of the neuromuscular junction, resulting in sustained muscle activation and eventually paralysis. Aldicarb-induced paralysis assay is an easy and fast method to determine whether synaptic transmission of a *C. elegans* mutant of interest is altered. This assay is based on the correlation of the rate of neurotransmitter release with the rate of paralysis. In this protocol, we describe a method for simultaneously assessing the aldicarb sensitivity of animals with different genotypes.

Pharmacological profiles of the GABA and acetylcholine receptors from the nematode, *Ascaris suum*.

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The pharmacological profiles of the acetylcholine and GABA receptors of *Ascaris* muscle have been investigated. Acetylcholine excites the muscle through activation of a nicotinic receptor which resembles the mammalian ganglionic receptor. DMPP is a potent agonist and alpha-bungarotoxin is a weak antagonist. GABA inhibits the muscle through activation of a chloride-linked receptor which in terms of agonist profile resembles the mammalian GABA-A receptor. However, bicuculline is inactive and picrotoxin is a very weak antagonist. Avermectin acts as a non-competitive antagonist at this GABA receptor with an IC-50 value in the low microM range.

BOTOX: Broadening the Horizon of Dentistry.

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Botox has been primarily used in cosmetic treatment for lines and wrinkles on the face, but the botulinum toxin that Botox is derived from has a long history of medically therapeutic uses. For nearly 13 years, until the introduction of Botox Cosmetic in 2002, the only FDA-approved uses of Botox were for crossed eyes (strabismus) and abnormal muscle spasms of the eyelids (blepharospasm). Since then botulinum A, and the seven other forms of the botulinum toxin, have been continuously researched and tested. Botox is a neurotoxin derived from bacterium *Clostridium botulinum*. The toxin inhibits the release of acetylcholine (ACH), a neurotransmitter responsible for the activation of muscle contraction and glandular secretion, and its administration results in reduction of tone in the injected muscle. The use of Botox is a minimally invasive procedure and is showing quite promising results in management of muscle-generated dental diseases like Temporomandibular disorders, bruxism, clenching, masseter hypertrophy and used to treat functional or esthetic dental conditions like deep nasolabial folds, radial lip lines, high lip line and black triangles between teeth.

Differences in time to peak carbachol-induced contractions between circular and

longitudinal smooth muscles of mouse ileum.

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The muscular layer in the GI tract consists of an inner circular muscular layer and an outer longitudinal muscular layer. Acetylcholine (ACh) is the representative neurotransmitter that causes contractions in the gastrointestinal tracts of most animal species. There are many reports of muscarinic receptor-mediated contraction of longitudinal muscles, but few studies discuss circular muscles. The present study detailed the contractile response in the circular smooth muscles of the mouse ileum. We used small muscle strips (0.2 mm × 1 mm) and large muscle strips (4 × 4 mm) isolated from the circular and longitudinal muscle layers of the mouse ileum to compare contraction responses in circular and longitudinal smooth muscles. The time to peak contractile responses to carbamylcholine (CCh) were later in the small muscle strips (0.2 × 1 mm) of circular muscle (5.7 min) than longitudinal muscles (0.4 min). The time to peak contractile responses to CCh in the large muscle strips (4 × 4 mm) were also later in the circular muscle (3.1 min) than the longitudinal muscle (1.4 min). Furthermore, a muscarinic M2 receptor antagonist and gap junction inhibitor significantly delayed the time to peak contraction of the large muscle strips (4 × 4 mm) from the circular muscular layer. Our findings indicate that muscarinic M2 receptors in the circular muscular layer of mouse ileum exert a previously undocumented function in gut motility via the regulation of gap junctions.

Distribution and role of aspartate in the nervous system of the chaetognath *Sagitta*.

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Cholinergic control of locomotory muscles in chaetognaths is monitored by diffuse transmitter release through layers of collagen fibers that form the connective stratum of the hydroskeleton. Despite the lack of morphologically defined synaptic junctions, the control of locomotor activity in chaetognaths is highly specific and allows complex behavioral patterns. This complexity suggests the existence of neuromediators acting to modulate the effects of the main motor neurotransmitter, acetylcholine, on muscular contraction. Immunocytochemical investigations performed in *Sagitta friderici* by using antibodies directed against L-aspartate revealed the presence of the amino acid within abundant fiber networks regularly distributed in the head, trunk, and tail and within discrete groups of cell bodies. In addition to known components of the sensory and motor nervous systems, L-aspartate immunoreactivity revealed previously undescribed intraepidermal networks of axonal profiles. With the exception of two giant anterior fibers radiating from the ventral ganglion, L-aspartate-immunoreactive processes were usually thin and varicose, occasionally making an anastomosis. As indicated by electron microscopy, L-aspartate-immunoreactive varicosities apposed to the connective stratum were filled with synaptic-like vesicles but displayed no synaptic differentiation. Physiologic investigations suggested a potent inhibitory effect of L-aspartate on acetylcholine-induced muscle contraction. The wide distribution pattern of immunoreactive profiles suggests an important role of L-aspartate in motor and sensory functions in chaetognaths. Although classified among excitatory amino acids in vertebrates, aspartate may function as an inhibitory modulator of acetylcholine-induced muscle contraction in these enterocoelous gastroneurians.

Detailed characterization of alterations of chromosomes 7, 9, and 10 in glioblastomas

as assessed by single-nucleotide polymorphism arrays.

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Glioblastomas are cytogenetically heterogeneous tumors that frequently display alterations of chromosomes 7, 9p, and 10q. We used high-density (500K) single-nucleotide polymorphism arrays to investigate genome-wide copy number alterations and loss of heterozygosity in 35 primary glioblastomas. We focused on the identification and detailed characterization of alterations involving the most frequently altered chromosomes (chromosomes 7, 9, and 10), the identification of distinct prognostic subgroups of glioblastomas based on the cytogenetic patterns of alteration for these chromosomes, and validation of their prognostic impact in a larger series of tumors from public databases. Gains of chromosome 7 (97%), with or without epidermal growth factor receptor (EGFR) amplification, and losses of chromosomes 9p (83%) and 10 (91%) were the most frequent alterations. Such alterations defined five different cytogenetic groups with a significant effect on patient survival; notably, EGFR amplification (29%) was associated with a better survival among older patients, as confirmed by multivariate analysis of a larger series of glioblastomas from the literature. In addition, our results provide further evidence about the relevance of other genes (eg, EGFR, CDKN2A/B, MTAP) in the pathogenesis of glioblastomas. Altogether, our results confirm the cytogenetic heterogeneity of glioblastomas and suggest that their stratification based on combined assessment of cytogenetic alterations involving chromosomes 7, 9, and 10 may contribute to the prognostic evaluation of glioblastomas.

Diagnostic challenges in complicated case of glioblastoma.

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Glioblastoma is the commonest primary malignant brain tumor, with a very poor prognosis and short overall survival. It is characterized by its high intra- and intertumoral heterogeneity, in terms of both the level of single-nucleotide variants, copy number alterations, and aneuploidy. Therefore, routine diagnosis can be challenging in some cases. We present a complicated case of glioblastoma, which was characterized with five cytogenomic methods: interphase fluorescence in situ hybridization, multiplex ligation-dependent probe amplification, comparative genomic hybridization array and single-nucleotide polymorphism, targeted gene panel, and whole-genome sequencing. These cytogenomic methods revealed classical findings associated with glioblastoma, such as a lack of IDH and TERT mutations, gain of chromosome 7, and loss of chromosome 10. At least three pathological clones were identified, including one with whole-genome duplication, and one with loss of 1p and suspected loss of 19q. Deletion and mutation of the TP53 gene were detected with numerous breakends on 17p and 20q. Based on these findings, we recommend a combined approach to the diagnosis of glioblastoma involving the detection of copy number alterations, mutations, and aneuploidy. The choice of the best combination of methods is based on cost, time required, staff expertise, and laboratory equipment. This integrated strategy could contribute directly to tangible improvements in the diagnosis, prognosis, and prediction of the therapeutic responses of patients with brain tumors.

Mitochondrial DNA copy number as a prognostic marker is age-dependent in adult glioblastoma.

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Glioblastoma (GBM) is the most common and aggressive form of glioma. GBM frequently displays chromosome (chr) 7 gain, chr 10 loss and/or EGFR amplification (chr7+/chr10-/EGFRamp). Overall survival (OS) is 15 months after treatment. In young adults, IDH1/2 mutations are associated with longer survival. In children, histone H3 mutations portend a dismal prognosis. Novel reliable prognostic markers are needed in GBM. We assessed the prognostic value of mitochondrial DNA (mtDNA) copy number in adult GBM. mtDNA copy number was assessed using real-time quantitative PCR in 232 primary GBM. Methylation of POLG and TFAM genes, involved in mtDNA replication, was assessed by bisulfite-pyrosequencing in 44 and 51 cases, respectively. Median age at diagnosis was 56.6 years-old and median OS, 13.3 months. 153/232 GBM (66 %) displayed chr7+/chr10-/EGFRamp, 23 (9.9 %) IDH1/2 mutation, 3 (1.3 %) H3 mutation and 53 (22.8 %) no key genetic alterations. GBM were divided into two groups, "Low" (n = 116) and "High" (n = 116), according to the median mtDNA/nuclear DNA ratio (237.7). There was no significant difference in OS between the two groups. By dividing the whole cohort according to the median age at diagnosis, OS was longer in the "High" vs "Low" subgroup (27.3 vs 15 months, P = .0203) in young adult GBM (n = 117) and longer in the "Low" vs "High" subgroup (14.5 vs 10.2 months, P = .0116) in older adult GBM (n = 115). POLG was highly methylated, whereas TFAM remained unmethylated. mtDNA copy number may be a novel prognostic biomarker in GBM, its impact depending on age.

Comparative genomic hybridization of human malignant gliomas reveals multiple amplification sites and nonrandom chromosomal gains and losses.

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Nine human malignant gliomas (2 astrocytomas grade III and 7 glioblastomas) were analyzed using comparative genomic hybridization (CGH). In addition to the amplification of the EGFR gene at 7p12 in 4 of 9 cases, six new amplification sites were mapped to 1q32, 4q12, 7q21.1, 7q21.2-3, 12p, and 22q12. Nonrandom chromosomal gains and losses were identified with overrepresentation of chromosome 7 and underrepresentation of chromosome 10 as the most frequent events (1 of 2 astrocytomas, 7 of 7 glioblastomas). Gain of a part or the whole chromosome 19 and losses of chromosome bands 9pter-23 and 22q13 were detected each in five cases. Loss of chromosome band 17p13 and gain of chromosome 20 were revealed each in three cases. The validity of the CGH data was confirmed using interphase cytogenetics with YAC clones, chromosome painting in tumor metaphase spreads, and DNA fingerprinting. A comparison of CGH data with the results of chromosome banding analyses indicates that metaphase spreads accessible in primary tumor cell cultures may not represent the clones predominant in the tumor tissue.

Accumulation of chromosomal changes in human glioma progression. A cytogenetic study of 50 cases.

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Cytogenetic studies of 50 human gliomas, including three oligodendrogliomas, 16 grade I-III astrocytomas, and 31 glioblastomas multiforme, were performed using the short-term tissue culture method. The most common numerical chromosome aberrations were +7, -9, -10, -14, and loss of a sex

chromosome. Structural changes involved predominantly the following chromosome arms: 1q, 2q, 6q, 7q, 9p, 14q, 17p, and 18p. Losses of chromosomes 9, 10, and 14, often occurring simultaneously and in polyploid clones, were observed almost exclusively in high-grade gliomas, and appear to constitute important events during glioma progression.
